

ME416

Ground Vehicle Dynamics

Tires, Lateral and Longitudinal Forces

Fall 2018

[1]

Introduction

- The tire is the interface between the vehicle and the ground and as such all performance, including stability, of a ground vehicle originates from the response of the tire
 - The tires must generate lateral force to allow for cornering
 - The tires must generate longitudinal forces to allow for braking and acceleration
 - A fundamental component of pneumatic tires is the generation of aligning moments and rolling resistance

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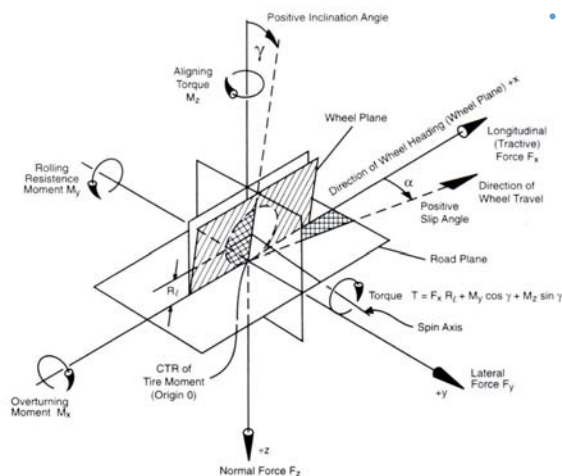
Introduction (2)

- Pneumatic tires generate their forces by a variety of mechanisms
 - Pneumatic tires can be partially described with dry friction assumptions for a very limited (and incomplete) portion of their operating range
 - Pneumatic tires also tend to grip or “gear” into the surface texture of the ground to produce additional grip – the mechanics of this behavior are not very well understood
 - Pneumatic tires generate their forces with a varying combination of “dry friction” and “gear” grip up until they exceed the effective coefficient of friction and begin to slide – unfortunately, the mechanisms that govern how effective sliding coefficient of friction is created is not well understood either

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Lateral Force SAE Convention

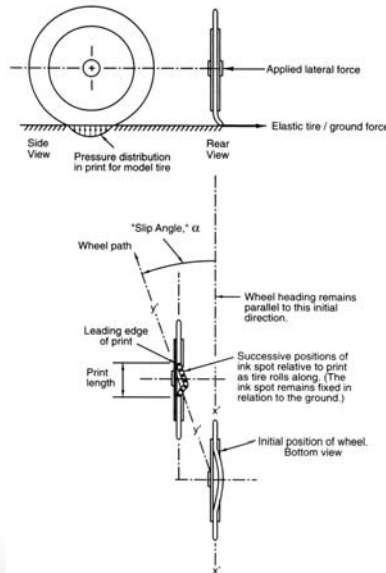


- According to SAE J670e, lateral tire force
 - Originates at the intersection of the wheel center plane with the ground plane at a point directly below the center of the upright of the wheel – we will generally refer to this as the center of the tire contact patch
 - Lies in the horizontal road plane
 - Perpendicular to the direction the wheel is headed assuming there is no inclination or camber exists

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Lateral Force Generation - Static

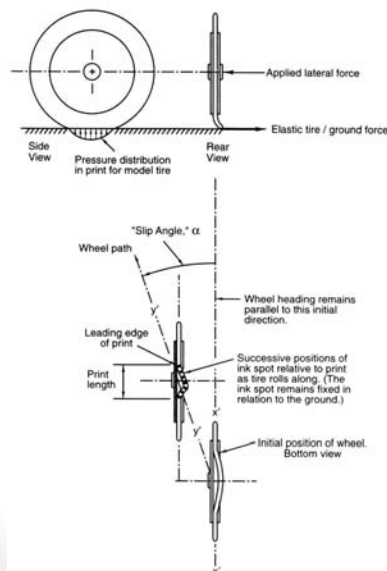


- Consider a simple model tire consisting of a sheet of hard rubber sandwiched between two metal disks
- Static lateral stiffness and sliding
 - If the model is pressed lightly onto a surface and then pushed sideways we get a pressure distribution and deformation as shown at top
- Consider
 - Increasing the vertical load
 - Increasing the lateral load
 - Increasing vertical and lateral load
- What happened?
 - Spring like behavior, vertically and laterally
 - Tire slides laterally at some point and the lateral force remains relatively constant thereafter

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Lateral Force Generation – Rolling

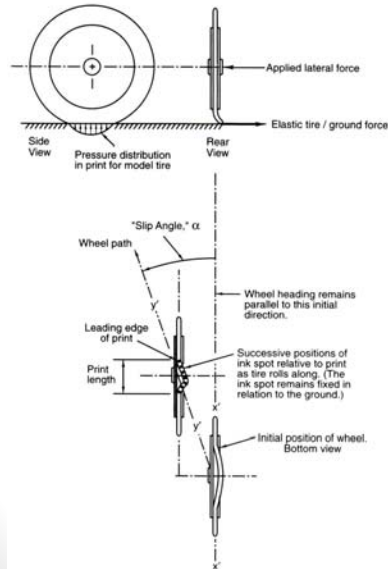


- With the same simple model tire consisting of a sheet of hard rubber sandwiched between two metal disks
- Rolling
 - Again, the model is pressed lightly onto a surface and then pushed sideways
 - At the same time the wheel is rolled along
 - The wheel will move at an angle to the wheel plane (heading)
 - This angle is called the Slip Angle
 - Note there is no sliding or slipping for this low lateral force case

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Lateral Force Generation – Rolling (2)

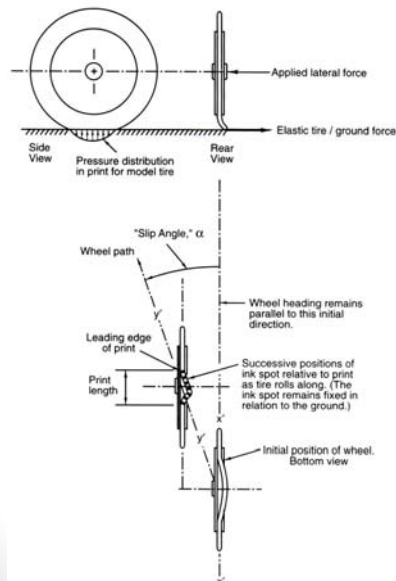


- The lateral distortion of the tire is produced as the tire/model comes into contact with the ground
- There is no slip of each contact point relative to the ground
 - Therefore, as the tire continues to move, i.e. the spot in contact with the ground moves aft in the print, the lateral deflection relative to the wheel plane increases
- This continues until near the end of the contact print where
 - The local vertical load is reduced
 - The local friction force is reduced due to the reduced vertical load
 - The friction force is too low to produce the lateral deflection and the deflection reduces quickly to zero at the end of the print

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Lateral Force Generation – Rolling (3)



- The previous description was for low slip angles
- At higher lateral loads higher slip angles will result
 - The tire is distorted further
 - The onset of actual sliding of slipping will move forward from the rear of the print
- At some point, the lateral load is too great and the entire print is sliding
- When the entire print is sliding the lateral force generated is determined by the sliding coefficient of friction

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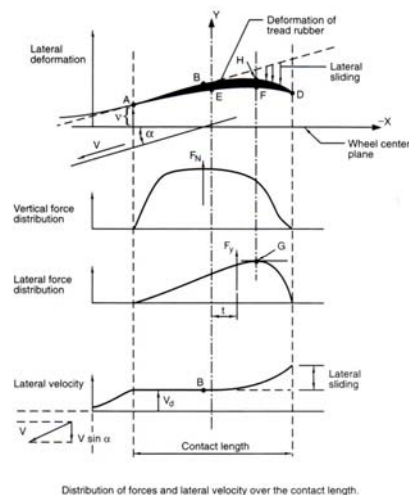
Lateral Force and Slip Angle

- When is lateral force generated as a result of slip angle?
- When is a slip angle generated as a result of lateral force?
- Consider the following cases
 - The front wheels are steered, a slip angle is created and a lateral force is generated causing the vehicle to yaw
 - On a tire test machine the a tire is set a slip angle and the lateral force generated is recorded
 - A sidewind strikes a vehicle and the sideforce is reacted by tire lateral force due to changes in tire slip angle
 - When cornering or on a banked road the lateral force is reacted by changes in tire slip angles and resultant tire forces

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Tire Contact Patch – Lateral Force

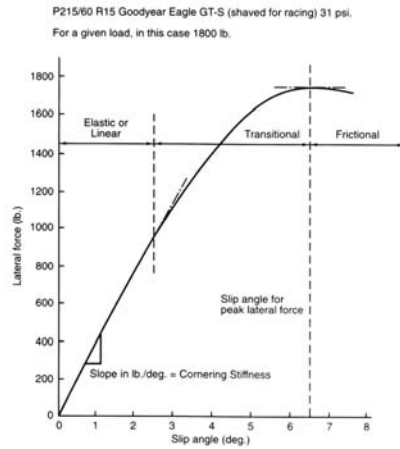


- Representative results on a real tire (ref: Gough) confirm the behavior of real tires as similar to the simple model we considered
- Some observations:
 - The data shown is for low speed tests (1-2mph)
 - The centroid of the lateral force is aft of the center point of the patch
 - This distance is the pneumatic trail
 - The pneumatic trail times the lateral force is the aligning moment
 - Lateral slippage of the rubber occurs at the back of the patch
 - Tire lateral force characteristics in the elastic range (small slip angles) are largely independent of speed

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Tire Lateral Force Data

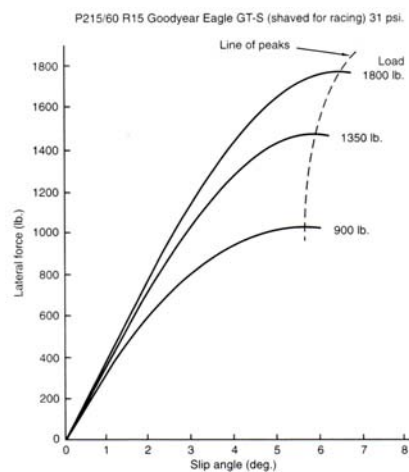


- The low speed tests are acceptable in the elastic range of the tire, i.e. low slip angles
- At higher slip angles, high speed testing is required
- A typical result for a high speed test is presented at right
 - Test data is for a single normal loading
 - Three distinct regions of tire behavior
 - Elastic
 - Transitional
 - Frictional
 - In the elastic region we can define the Cornering Stiffness as the slope of the lateral force vs. slip angle curve

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Tire Lateral Force Data (2)

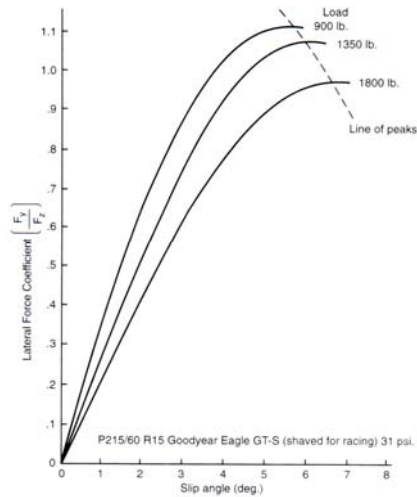


- At right is the same tire tested at different normal loadings
- Note that as the load increases
 - The increase in lateral force does not increase at the same rate
 - The peak lateral force generated is generally at a higher slip angle

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Tire Lateral Force Data (3)



- We can normalize this data to give a dimensionless measure of the lateral force obtained in relation to the normal load

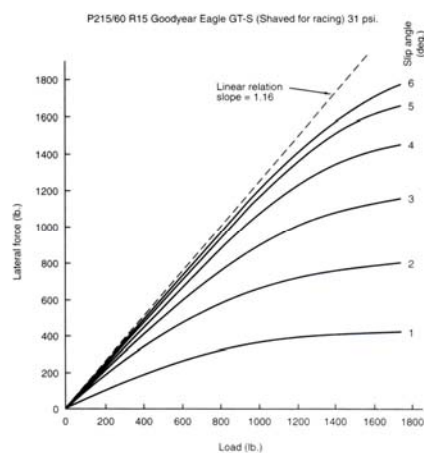
$$\text{Lateral Force Coefficient} = \frac{\text{Lateral Force}}{\text{Normal Load}} = \frac{F_y}{F_z}$$

- Again, we can see the fall off of lateral force coefficient as the normal load increases
- This is the Tire Load Sensitivity
- The lateral friction coefficient is approximately independent of speed (neglecting aerodynamic loads) and is most directly influenced by tire compound

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Elastic, Transitional and Friction



- Tire load sensitivity has a major effect on balance behavior at the traction limit
 - As the normal load varies due to cornering, acceleration and braking the response of the tire is also constantly changing
- An alternate way of looking at the previous tire data is presented to the right
 - This behavior is typical for modern radial tires
 - All curves are non-linear
 - All curves show a reduction in the lateral force coefficient at higher normal loads

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Elastic, Transitional and Friction (2)

- The tire response in the transitional zone is the result of many tire design factors
 - As the tire reaches its traction limit it may 'let go' gradually or abruptly
 - Tires are developed with specific traits in the transition regime depending on their intended use
 - Road cars will generally have very gradual and benign behavior in the transition zone often at the detriment of peak grip
 - Traditionally, the higher the performance capability of the tire the more abruptly the tire would lose traction – note this is not because it's desirable but because of an (assumed) higher level of capability in the driver and therefore an acceptable tradeoff
 - This is a major area of development in modern road racing tires

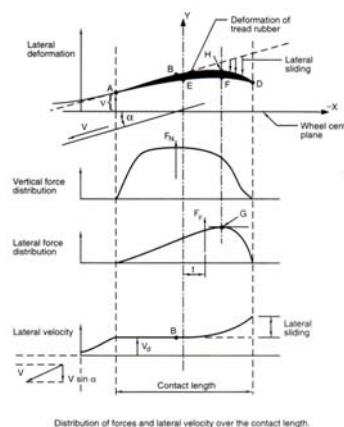
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Aligning Torque

- Recall the shape of the contact patch
- The contact patch due to lateral distortion is roughly triangular (in the linear region) with the lateral distortion increasing from front to rear
- The point of application of the resultant lateral force is located behind the center of the wheel, this is called the pneumatic trail
- Where,

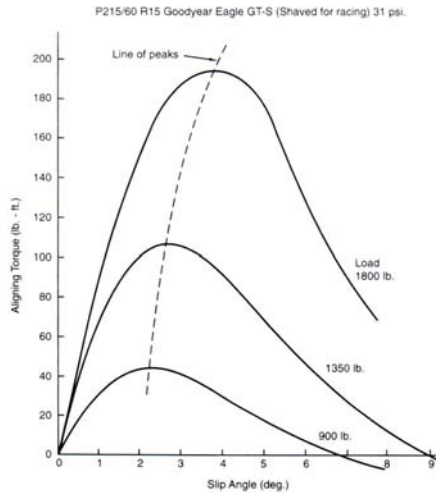
$$\text{Aligning Torque} = \text{Pneumatic Trail} * \text{Resultant Lateral Force}$$



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Aligning Torque (2)



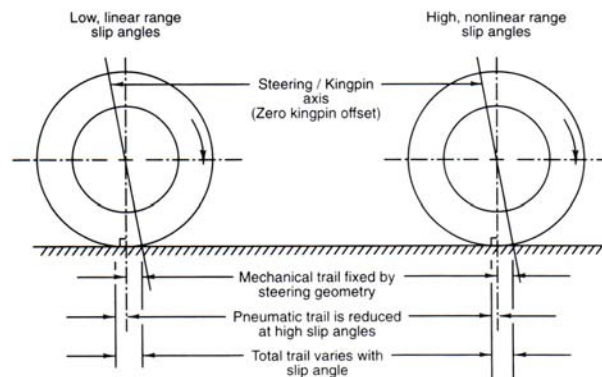
- At low slip angles the high stresses in the aft part of the contact patch tend to return the wheel to straight ahead
- At high slip angles the aft part of the contact patch is starting to slip and therefore has a smaller contribution to the total lateral force produced
 - As the aft contribution percentage is reduced the pneumatic trail is reduced
 - As pneumatic trail is reduced the aligning torque is also reduced
 - When the tire transitions to full sliding the aligning torque is zero or may even be negative

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Mechanical Trail, Pneumatic Trail and Steering Torques

- We have been discussing trail in terms of the tire (pneumatic) response
- Most cars are also designed with mechanical trail as well
- The sum of the mechanical and pneumatic trail times the lateral force gives the steering torque about the steering axis

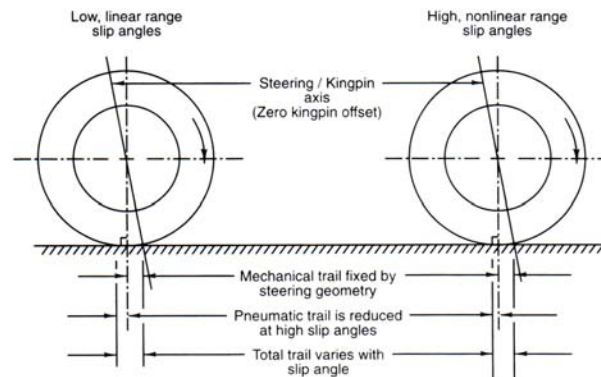


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Mechanical Trail, Pneumatic Trail and Steering Torques (2)

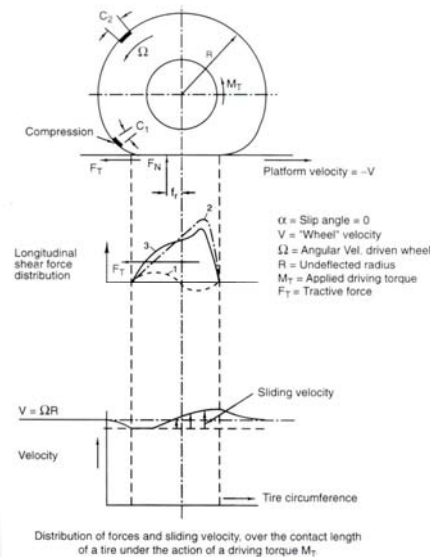
- Ultimately, the steering torque is provided/reacted at the steering wheel rim by the driver
- The torque at the steering wheel rim is a major component of driver feedback of grip level and impending grip limit
- Additionally, the level of torque the driver must provide must also be managed by geometry, tire selection or power assistance



Longitudinal Forces

- In addition to lateral forces the tires must also provide longitudinal forces
 - Braking
 - Tractive
- These forces are generated in a similar manner to lateral forces
 - The forces result from deformation of the tire
- However, braking and tractive behavior are not necessarily equivalent
 - We will consider each independently

Contact Patch – Tractive Force

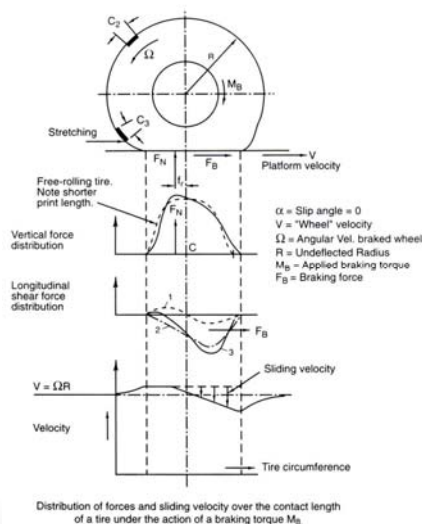


- The applied driving torque (M_T) produces a tractive force (F_T) from the road to the tire
- The tractive force displaces the contact patch forward relative to the axle
- The deformation of the contact patch is propagated into the circumference
 - The length of a given segment is compressed as it approaches the contact patch
 - The shear forces build as the segment travels aft in the contact patch yielding curve 2
- A free rolling tire also creates longitudinal stresses in the contact patch yielding curve 1
- The total shear force distribution is a combination (approximately) of curve 1 and 2 represented as curve 3

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Contact Patch – Braking Force



- The applied braking torque (M_B) produces a braking force (F_B) from the road to the tire
- The braking force displaces the contact patch aft relative to the axle
- The deformation of the contact patch is propagated into the circumference
 - The length of a given segment is compressed aft of the contact patch and tends to stretch the segments forward of the contact patch
 - The shear forces build as the segment travels aft in the contact patch yielding curve 2
- A free rolling tire also creates longitudinal stresses in the contact patch yielding curve 1
- The total shear force distribution is a combination of curve 1 and 2 represented as curve 3

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Force Transfer Between Tire and Road

- Due to the viscoelastic properties of the tire there will be a slip as soon as the tire turns and a torque is transmitted
- Tractive slip str means that the circumferential velocity of the wheel, v_w is greater than the vehicle, v_{veh}

$$s_{tr} = \frac{v_w - v_{veh}}{v_w}$$

with

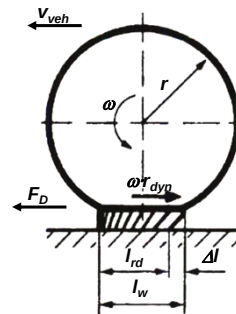
$$v_w = \omega * r_{dyn} = 2 * \pi * n_w * r_{dyn}$$

- Slip results in a loss in power

$$P_{s,tr} = F_D * s_{tr} * v_w$$

- Braking slip is similar with the circumferential velocity of the wheel, v_w less than the vehicle, v_{veh}

$$s_{tr} = \frac{v_{veh} - v_w}{v_{veh}}$$



Tire model for slip („brush model“)

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Traction/Braking Force vs. Slip Ratio

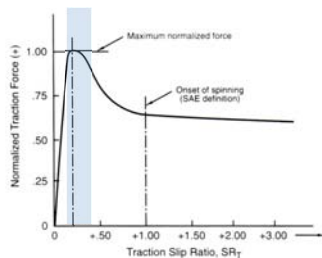
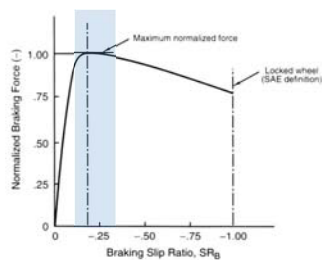


Figure 2.16 Typical traction—slip ratio curve; slip angle = 0°.



- Typical traction and braking curves are presented to the right on a dry surface
- Tractive force (F_T) and braking force (F_B) are a function of slip ratio
- Traction/Braking increases quickly as the slip ratio increases up to the peak
 - The peak for traction is actually effectively pretty broad for range of slip ratios of ~10%-30% depending on tire...
 - For braking it is similar with less aggressive a drop-off at higher slip ratios
- An increase in slip gives an increased force which damps the rotational velocity
- At the peak the system is unstable and the wheel tends to spin-up (traction) or lock-up (braking)

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What About Different Road Conditions?

- For the adhesion coefficient or normalized longitudinal force we observe a typical variation of 'grip' with road conditions
- Note that this will vary with road surface, condition of tire, tire and tread type etc.

